

4.2

Data Acquisition

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OBJECTIVE

This exercise shows how transactional data can be loaded into the data acquisition layer of the Data Warehouse. This step is also known as Extract step of the ETL process. Acquiring the data will serve as a basis for modeling the data in further case studies.

ACTIVITIES

Learn how to load sales transaction data from a source system in the data acquisition layer.

SOFTWARE PREREQUISITES

- Eclipse with SAP BW Modeling Tools
- SAP GUI

DATA SET

Global Bike (GB) data set

SYSTEM ACCESS

SAP BW/4HANA 2.0 from your UCC

Scenario

In this exercise Jan will load the *sales transaction* data (transactional data) from the source system into the data acquisition layer of the Data Warehouse for Global Bike Germany and US.

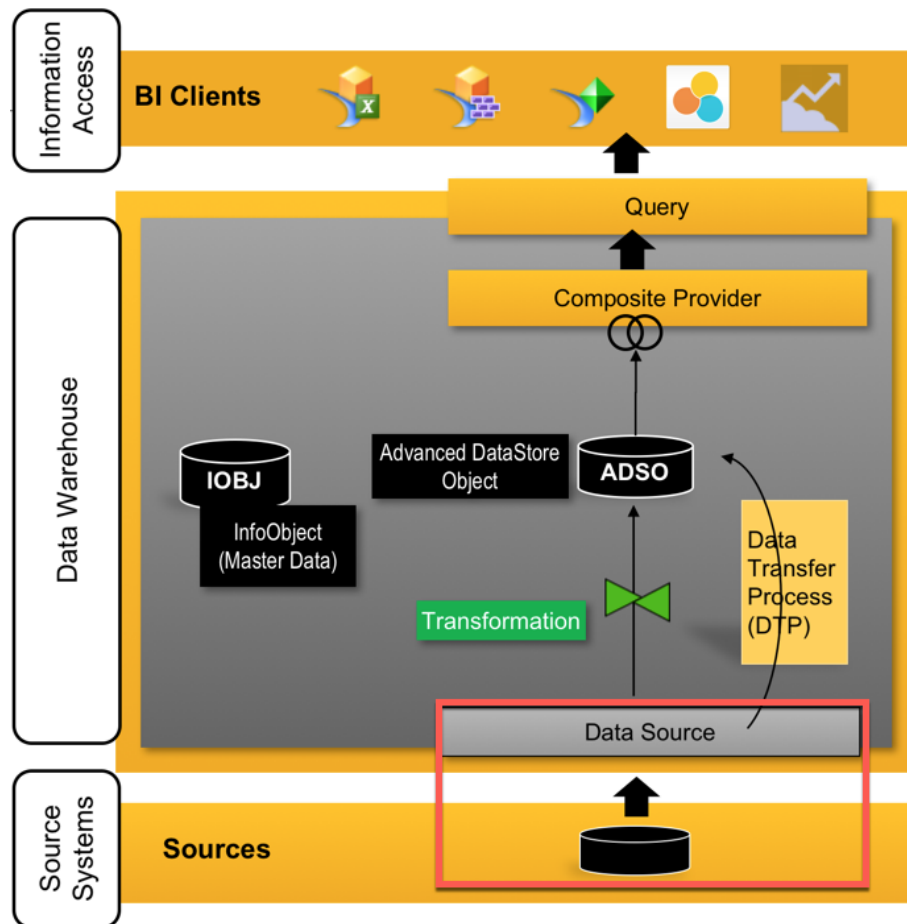


Figure 1

First you need to understand how the sales transaction data from the source system looks like. Then you will create a DataSource as a preparation for the loading process into the data acquisition layer of SAP BW4/HANA 2.0. Once you do this for the sales transaction data from Global Bike Germany you will repeat this for the US.

The graphic below displays the complete process (4 tasks).

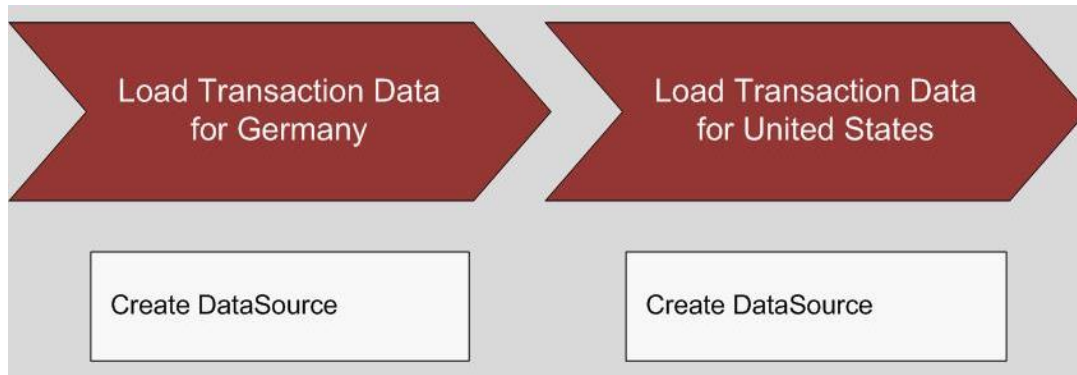


Figure 2

1. Create a DataSource for the sales transaction data for Global Bike Germany.

In this step Jan prepares for the loading process into the data acquisition layer of BW4/HANA.

- a. Your instructor will provide you with the sales transaction data for GB Germany and GB US as CSV files.
- b. **Download** these files.
- c. Open the CSV file *Sales Data DE.csv* (use a text editor like **Notepad**, **NOT Excel**!)
Explore the structure of the data. Note the:
 - i. Header lines
 - ii. Separator between two columns
 - iii. Number format (decimal & thousand separator, varies by country)
 - iv. Date format

Additional Information

The *DataSource* is the object that represents data of source systems. A DataSource in BW consists of a set of fields and several adapter settings specific for the type of source system. The fields in the DataSource contain e.g. information about the data type.

- d. Open **Eclipse**.
- e. Open your Project.
- f. Open the **DataSource** tree and choose the source system **PC_FILE**.

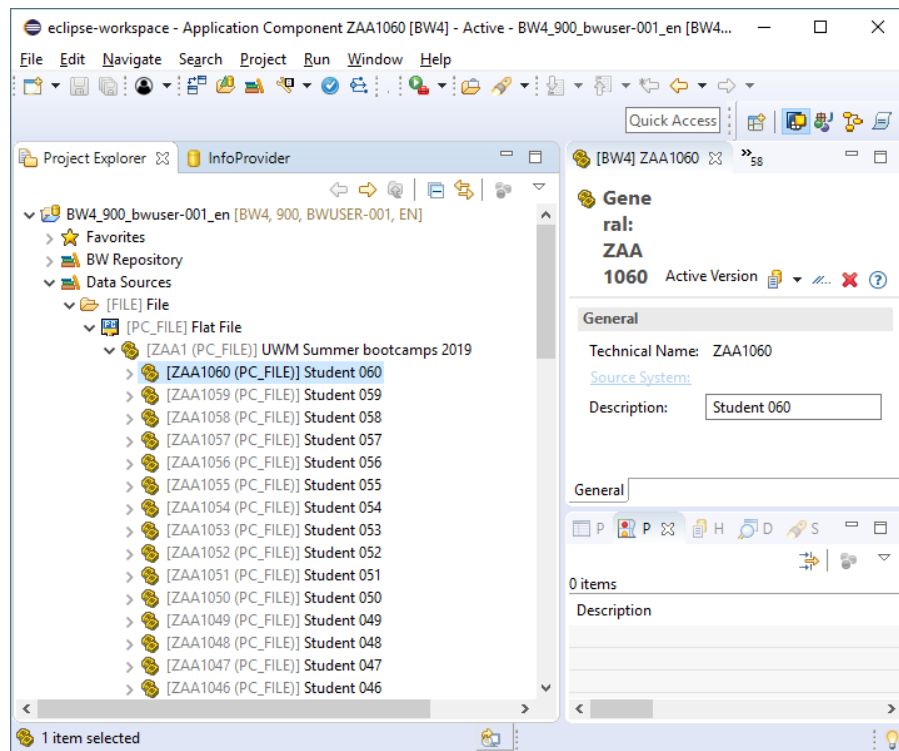


Figure 3

- g. Navigate to your own application component folder `ZXXX` → `ZXXX###` (name: `Student ###`) and choose **New** → **DataSource** in the context menu (right-click).

Note: When working with the exercise, please replace in all names XXX with the Id of your course, and ### with your user number (three digits) provided by your instructor.

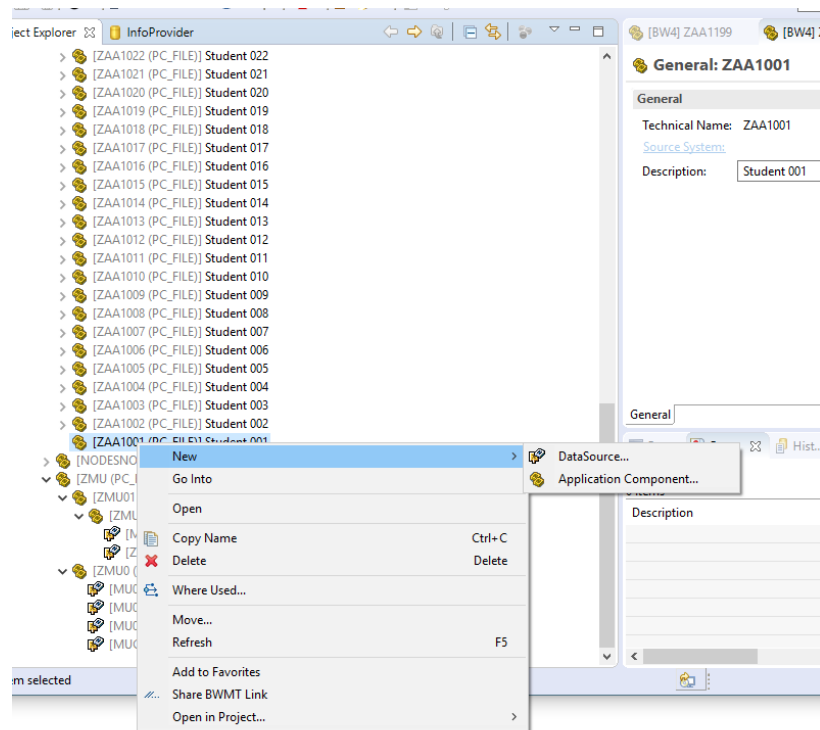


Figure 4

- h. Click **Next** → **Next**.
- i. On the next screen, create a DataSource **XXXDSDE###** (please replace **XXX** with the ID of your course and **###** with your number) of type *transactional data*.
- j. Enter description, **XXX DataSource Sales Data DE ###**.

New DataSource

Confirmation Page

BW Project: BW4_900_bwuser-001_en

Source System: PC_FILE

Source System Type: FILE

Application Component: ZAA1001 Browse...

☐ Add to Favorites

Name: AA1DSDE001

Description: AA1 Data Source Sales Data DE 001

DataSource Type: Transactional Data

? < Back Next > Finish Cancel

Figure 5

- k. **Finish.**
- l. The **Overview** screen shows the DataSource editor with several tabs.
- m. On tab **Extraction** choose the following settings:
 - i. Direct Access: **Not Allowed**
 - ii. Adapter: **Load text-type file from local workstation**
 - iii. File Name: **(the path to your CSV file, *Sales Data DE.csv*)**
 - iv. Ignore Header Rows: **1**
 - v. Data Format: **Separated with Separator (for example, CSV)**
 - vi. Data Separator: **;**
 - vii. Escape Sign: **"**
 - viii. Set Number format: **checked**
 - (1) Thousands separator: **none**
 - (2) Decimal Point separator: **,**

The screenshot shows the 'Extraction: AA1DSDE001' dialog box. It has a title bar with 'Active Version' and standard window controls. The dialog is divided into several sections:

- Extraction Properties:** Contains dropdowns for 'Delta Process' (set to 'No delta, only Full'), 'Direct Access' (set to 'Not Allowed'), 'Streaming' (set to 'Streaming cannot be triggered by this DataSource'), and 'Adapter' (set to 'Load Text-Type File from Local Workstation').
- Extractor-Specific Properties:** Includes a 'File Name' field with the path 'C:\Users\Nitin Kale\Google Drive\SAP\SAP BW4', an 'Ignore Header Rows' field set to '1', a 'System Codepage' field set to '4103 UTF-16LE Unicode / ISO/IEC 10646', and a checkbox for 'Specify Codepage to Replace System Codepage'. At the bottom are buttons for 'Show File Content...' and 'Derive Fields from File...'.
- Data Format:** Includes a 'Data Format' dropdown set to 'Separated by Separator (Example: CSV)', and fields for 'Data Separator' (set to ';') and 'Escape Sign' (set to '"'). Each has a 'Browse...' button and a 'Hex' checkbox.
- Conversion Properties:** Includes a checkbox for 'Set Conversion Language' and a checkbox for 'Set Number Format'. The 'Set Number Format' section has 'Thousands Separator' and 'Decimal Point Separator' fields.

Figure 6

- n. Click on **Save**.
- o. Click on **Derive Fields from File...**
- p. **Next**.

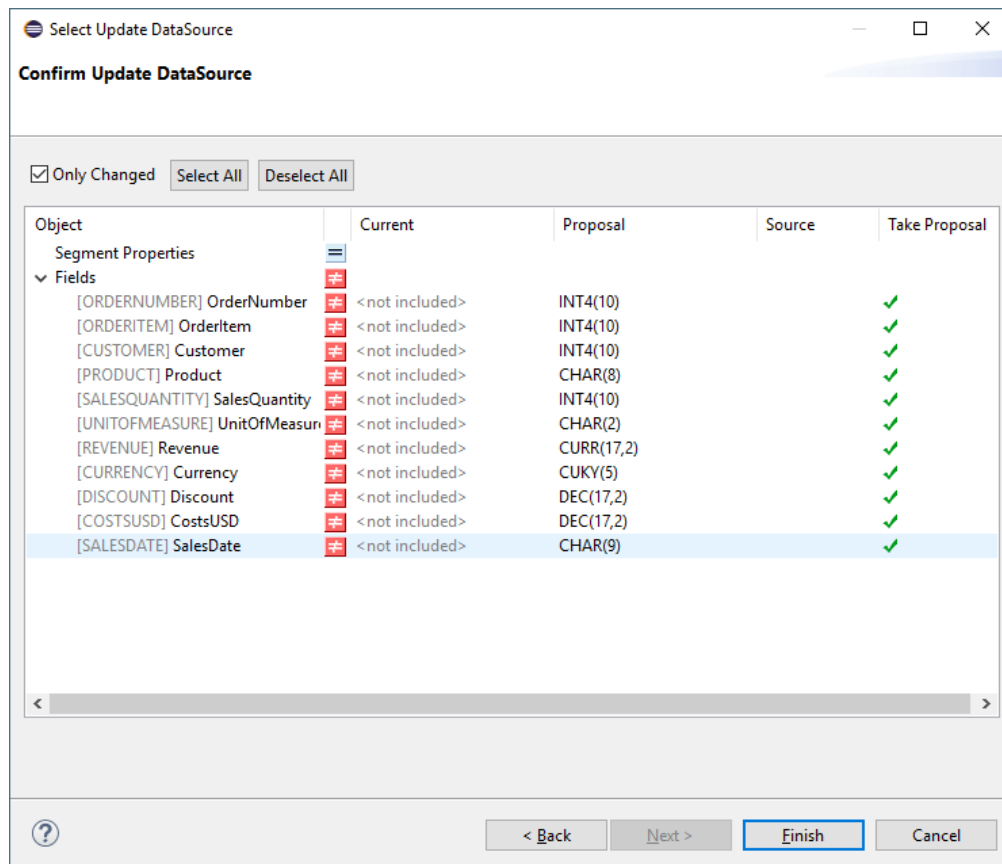


Figure 7

q. **Finish.**

r. We accept this proposal by clicking on the next tab **Fields**.

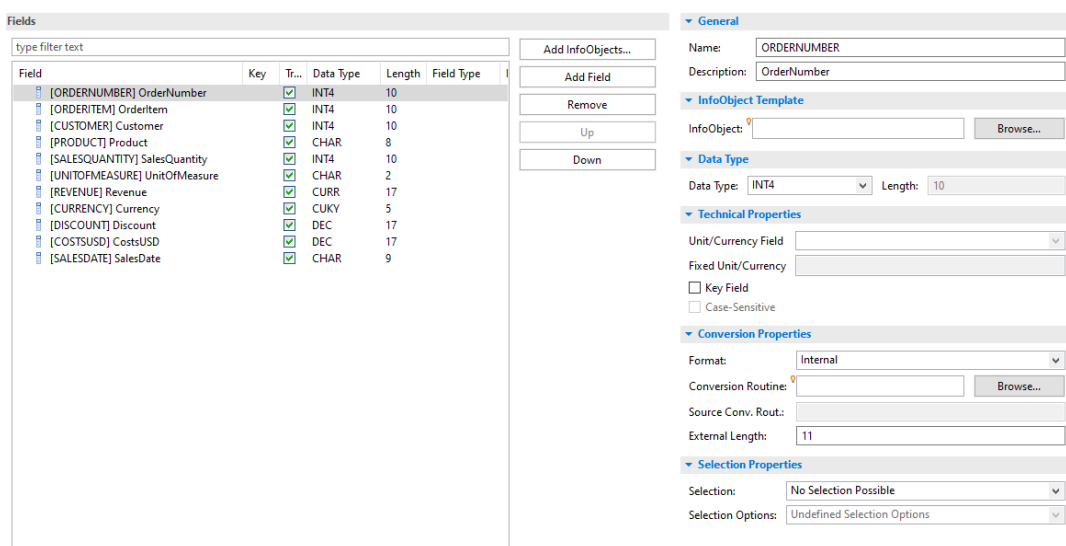


Figure 8

- s. On the tab **Fields** we will adjust the proposed type of some fields and other settings in the DataSource.

It is important that order of the fields matches the order of the fields in the file, so DO NOT CHANGE the order to the fields in the DataSource!

- t. Go to each Field and then change its properties on the right General panel.
- For the fields **ORDERNUMBER**, **ORDERITEM**, **CUSTOMER** select type **NUMC** and length **10**, **6**, **10**.
 - The field **PRODUCT** is **18** characters long (this is the length of the material number in SAP ERP)
 - SALESQUANTITY** has the type **QUAN** and the precision **15**. The corresponding **UNIT-field** is **UNITOFMEASURE**
 - REVENUE** & **DISCOUNT** are amount fields (type **CURR**), it is delivered in external format. Length is precision **17** with **2** (scale) decimal digits. Make sure that the reference to the currency field **CURRENCY** is set!
 - COSTSUSD** is also an amount field but is in fixed to currency **USD**. The type is **DEC** and no currency reference is necessary. Precision is **17** with **2** scale.
 - SALESDATE** is a date (type **DATS**), delivered in **External** format. This date has to be converted into the internal date format of SAP BW system e.g **20160530**. This is done with the conversion-exit **RSDAT**.
- u. Check the table below for the correct values. *Pay attention to all columns.*

Field	Description	Data Type	Length	Decimal	External	Conversion Routine	Format	Unit/Curr
OrderNumber	OrderNumber	NUMC	10		10		External	
OrderItem	OrderItem	NUMC	6		6		Internal	
Customer	Customer	NUMC	10		10		Internal	
Product	Product	CHAR	18		18		Internal	
SalesQuantity	SalesQuantity	QUAN	15		15		Internal	Unitof Measure
UnitofMeasure	UnitofMeasure	UNIT	3		3		Internal	
Revenue	Revenue	CURR	17	2	23		External	Currency
Currency	Currency	CUKY	5		5	RSCUK	External	
Discount	Discount	CURR	17	2	23		External	Currency
CostsUSD	CostsUSD	DEC	17	2	23		External	
SalesDate	SalesDate	DATS	8		10	RSDAT	External	

External/Internal: BW has its own internal format. If the incoming data from the source is not compliant with the internal format, you will get an error. So, we set it as External. In addition, we set a conversion routine. The routine will convert the external format to the internal format. When set to External, we can also choose the length of the incoming data.

Key figures of type CURR are always stored internally with:

- 2 decimal places
 - No thousand separator
 - The decimal point is .
- v. **Save**  and **activate** your DataSource.
- w. Now you can test your DataSource. Click **Show Data Preview** at the top right.
- x. Check the *Show data in internal format* option. The result should look like the following figure. Uncheck that option and observe the difference.

Data Preview for DataSource MU0DSDE (FILE)

Number of Records: 20 ☒ Show data in internal format

ORDERNUMBER	ORDERITEM	CUSTOMER	PRODUCT	SALESQUANTITY	UNITOFMEASURE	REVENUE	CURRENCY	DISCOUNT	COSTSUSD	SALESDATE
0000100001	000010	0000017000	PRTR1000	4	ST	10669.84	EUR	320.10	8320.00	20070101
0000100001	000020	0000017000	DXTR1000	8	ST	20005.92	EUR	600.18	15600.00	20070101
0000100001	000030	0000017000	DXRD2000	2	ST	2750.82	EUR	82.52	2046.00	20070101
0000100001	000040	0000017000	ORWN1000	5	ST	10419.75	EUR	312.59	7500.00	20070101
0000100002	000010	0000015000	PRTR1000	4	ST	10669.84	EUR	533.49	8320.00	20070103
0000100002	000020	0000015000	ORHT2000	7	ST	9919.60	EUR	495.98	7140.00	20070103
0000100002	000030	0000015000	SHRT1000	6	ST	150.06	EUR	7.50	90.00	20070103
0000100002	000040	0000015000	RHMT1000	1	ST	41.68	EUR	2.09	25.00	20070103
0000100002	000050	0000015000	FAID1000	21	ST	700.22	EUR	35.01	420.00	20070103
0000100002	000060	0000015000	PRRD1000	7	ST	23340.24	EUR	1167.01	17360.00	20070103
0000100002	000070	0000015000	ORHT2000	2	ST	2834.17	EUR	141.71	2040.00	20070103
0000100003	000010	0000022000	FAID1000	29	ST	966.98	EUR	29.01	580.00	20070103

Figure 9

- y. Please verify that all numbers and the sales date are displayed correctly.
- z. After activation you can find your DataSource in the navigation tree:

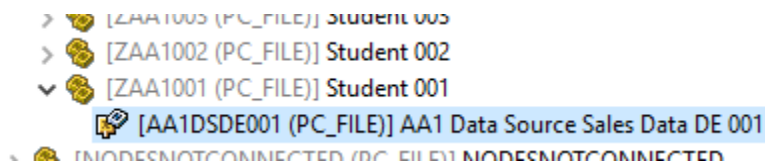


Figure 10

Question 1: What is a data source?

Question 2: You have created a data source for sales data. What do you think is the destination of this data in terms of the snowflake schema? Why?

Question 3: Go to the Data Preview and toggle the *Show data in internal format only* flag. Comment on the revenue formats when you toggle.

Question 4: What is the internal format for salesdate?

2. Create a DataSource for the sales transaction data for Global Bike US.

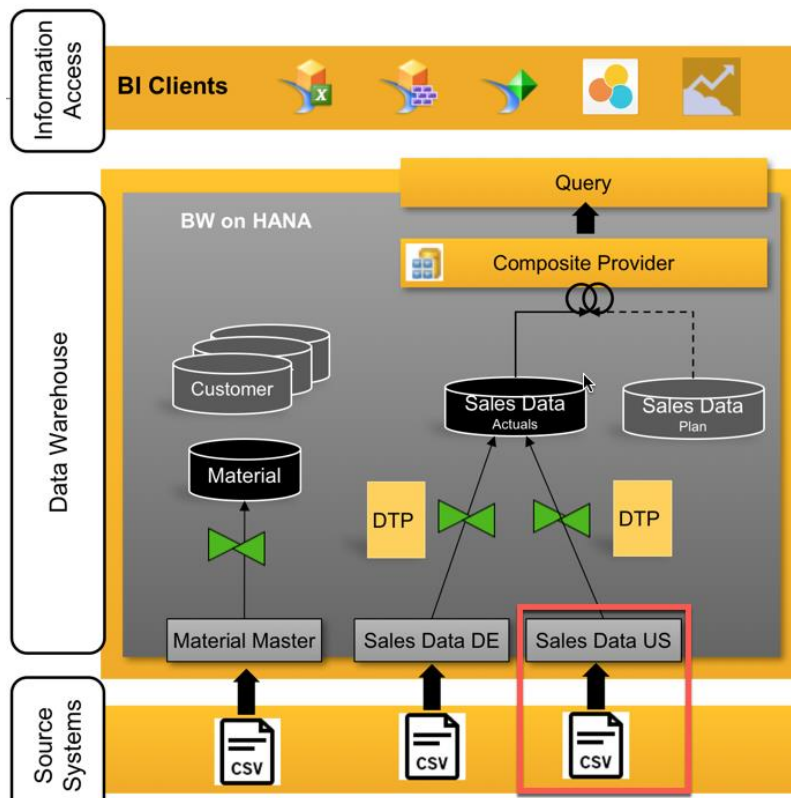
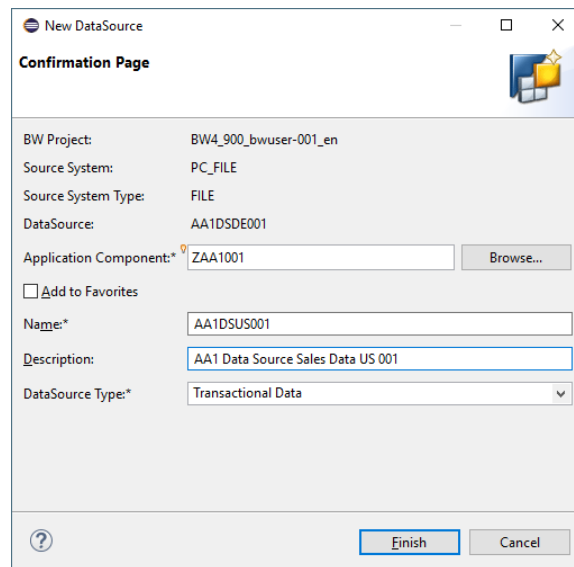


Figure 11

- Open the CSV file *Sales Data US.csv* with a text editor.

Question 5: What are the differences of US data as compared to the data from Germany?

- Because the data are different, we need to create a separate DataSource for US. However, because the structure of the file (the columns of the DataSource) are the same, we can copy the DataSource from step 1.
- Open **Eclipse**.
- Navigate to your DataSource `XXXDSDE###` (`XX` → `XXX` → `XXX###` (name: `Student ###`)) and choose **Copy** in the context menu (right-click).



New DataSource

Confirmation Page

BW Project: BW4_900_bwuser-001_en

Source System: PC_FILE

Source System Type: FILE

DataSource: AA1DSDE001

Application Component: ZAA1001 Browse...

☐ Add to Favorites

Name: AA1DSUS001

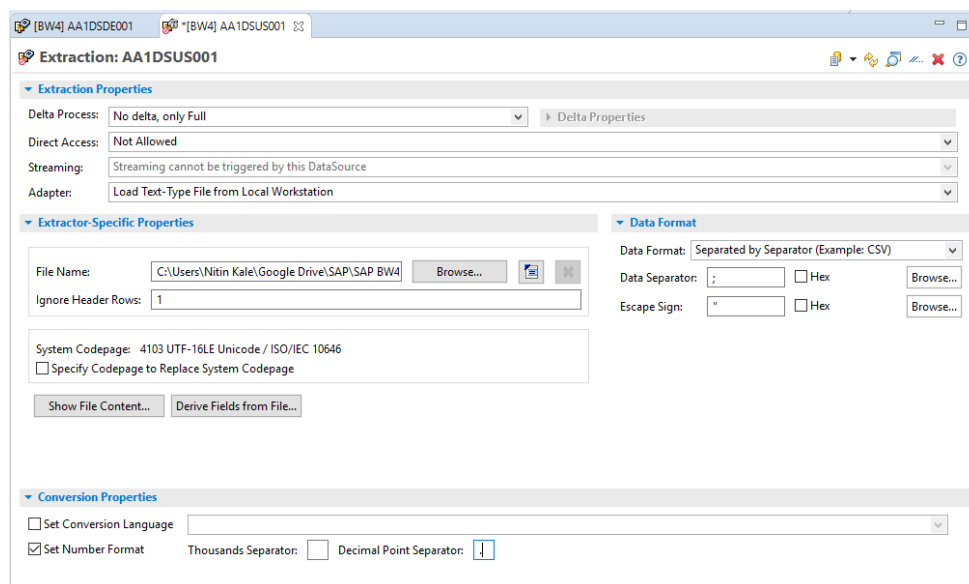
Description: AA1 Data Source Sales Data US 001

DataSource Type: Transactional Data

? Finish Cancel

Figure 12

- e. Choose as name **XXXDSUS###** for your copy.
- f. **Description:** **XXX DataSource Sales Data US ###**.
- g. **Finish**.
- h. On tab **Extraction**, change the settings for file name (browse to the *Sales Data US.csv*) and number format separator accordingly.
 - i. Data Separator: ;
 - ii. Set Number Format - Decimal Point Separator: .



Extraction: AA1DSUS001

Extraction Properties

Delta Process: No delta, only Full Delta Properties

Direct Access: Not Allowed

Streaming: Streaming cannot be triggered by this DataSource

Adapter: Load Text-Type File from Local Workstation

Extractor-Specific Properties

File Name: C:\Users\Nitin Kale\Google Drive\SAP\SAP BW4 Browse...

Ignore Header Rows: 1

System Codepage: 4103 UTF-16LE Unicode / ISO/IEC 10646

☐ Specify Codepage to Replace System Codepage

Show File Content... Derive Fields from File...

Data Format

Data Format: Separated by Separator (Example: CSV)

Data Separator: ; ☐ Hex Browse...

Escape Sign: * ☐ Hex Browse...

Conversion Properties

☐ Set Conversion Language

☒ Set Number Format

Thousands Separator: ☐ Decimal Point Separator: .

Figure 13

- i. On tab **Fields**, everything can be left unchanged, because the different date format is handled by the conversion exit automatically!
- j. **Save**  and **activate** your DataSource.
- k. Test your DataSource in Preview by clicking on **Show Data Preview** at the top left.
- l. Toggle the internal data format option.

Number of Records: 650

ORDERNUMBER	ORDERITEM	CUSTOMER	PRODUCT	SALESQUANTITY	UNITOFMEASURE	REVENUE	CURRENCY	DISCOUNT	COSTSUSD	SALESDATE
0000200001	000010	0000002000	DXRD1000	7	ST	11,900.00	\$	357.00	7,378.00	01/01/2007
0000200002	000010	0000005000	PUMP1000	7	ST	196.00	\$	9.80	98.00	01/02/2007
0000200002	000020	0000005000	ORHT2000	1	ST	1,700.00	\$	85.00	1,020.00	01/02/2007
0000200003	000010	0000011000	ORWN1000	7	ST	17,500.00	\$	525.00	10,500.00	01/02/2007
0000200003	000020	0000011000	FAID1000	18	ST	720.00	\$	21.60	360.00	01/02/2007
0000200003	000030	0000011000	PRRD1000	6	ST	24,000.00	\$	720.00	14,880.00	01/02/2007
0000200003	000040	0000011000	FAID1000	13	ST	520.00	\$	15.60	260.00	01/02/2007
0000200003	000050	0000011000	PUMP1000	3	ST	84.00	\$	2.52	42.00	01/02/2007
0000200003	000060	0000011000	ORHT1000	5	ST	8,000.00	\$	240.00	4,800.00	01/02/2007
0000200003	000070	0000011000	DXRD1000	9	ST	15,300.00	\$	459.00	9,486.00	01/02/2007
0000200003	000080	0000011000	DXTR3000	4	ST	12,000.00	\$	360.00	7,800.00	01/02/2007
0000200004	000010	0000002000	PUMP1000	9	ST	252.00	\$	7.56	126.00	01/04/2007
0000200004	000020	0000002000	DXTR3000	2	ST	6,000.00	\$	180.00	3,900.00	01/04/2007
0000200005	000010	0000005000	DXTR3000	6	ST	18,000.00	\$	900.00	11,700.00	01/04/2007
0000200005	000020	0000005000	RKIT1000	8	ST	256.00	\$	12.80	128.00	01/04/2007
0000200005	000030	0000005000	CAGE1000	8	ST	144.00	\$	7.20	72.00	01/04/2007
0000200005	000040	0000005000	PRRD2000	3	ST	12,600.00	\$	630.00	7,812.00	01/04/2007
0000200005	000050	0000005000	PRTR3000	8	ST	25,600.00	\$	1,280.00	16,640.00	01/04/2007

Figure 14

Question 6: Why did we have to create two data sources, one for DE and one for US?

Question 7: What is the unit of measure of sales quantity?

Question 8: Are any characteristics mapped to the data source fields yet? Why? Why not?

Question 9: How many sales transactions do we have for US? How many for DE?

Question 10: Insert screenshots of the preview of your data In the two datasources.